

Risk Management & Risk Analysis

1. What is Risk?

uncertain future event that may or may not happen, but if it happens, it can **negatively affect the project**.

Example

Suppose:

- A developer leaves the project.
- A new developer joins.
- The new developer uses a different technology.

Now the team must:

- Deliver same functionality
- In same time
- With changed technology

This creates **schedule risk**

Risk Exposure

Risk exposure measures **how dangerous a risk is**.

$$RE = P \times C$$

Where:

- P = Probability of occurrence
 - C = Cost/Consequence if it occurs
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2. Risk Categories

A. Project Risks

These **affect project planning**.

Effects:

- Schedule delays
- Increased cost
- Resource problems

B. Technical Risks

These **affect software development** itself.

Effects:

- Difficult implementation
- Reduced quality
- Technology failures

C. Business Risks

These **threaten business success**.

Effects:

- Product failure
- Financial loss
- Project cancellation

Business Risk Subcategories

Market Risk

- Building software nobody wants.

Strategic Risk

- Product no longer matches company goals

Sales Risk

- Sales team cannot explain or sell product.

Management Risk

- Senior management stops supporting project.

Budget Risk

- Project loses funding.
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3. Types of Risks

- › **Known Risks:** Can be identified before project starts.
 - › **Predictable Risks:** Can be predicted from previous projects.
 - › **Unpredictable Risks:** Very difficult to foresee.
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4. Risk Management

Risk Management is the process of:

- **Identifying** risks
- **Analyzing** risks
- **Reducing** risks
- **Monitoring** risks

Goal: **Deliver project successfully despite uncertainties.**



5. Seven Principles of Risk Management

1. Maintain Global Perspective

Consider:

- Entire system
- Entire business

Not only software.

2. Take Forward Looking View

- Think about future problems.
- Prepare contingency plans.

3. Encourage Open Communication

- Everyone should report risks freely.

4. Integrate Risk Management

- Risk management should be part of development process.

5. Continuous Process

- Risks change over time.
- New risks appear.
- Old risks disappear.

6. Shared Product Vision

- Everyone should understand project goals.

7. Encourage Teamwork

- Use knowledge of entire team
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6. Reactive vs Proactive Risk Management

Reactive Approach

Philosophy: "We'll handle it when it happens."

- project team reacts to risks when they occur
- Team doesn't actively try to prevent risks in advance
- **No preparation**
- **Action after failure**

Fix on Failure: Problem occurs → fix it.

Crisis Management: Project already in trouble.

Fire Fighting / mitigation: Extra resources added at last minute.

- Resources kept ready in anticipation of possible problem to lessen the negative impact
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Proactive Approach

Philosophy: Prevent problems before they occur.

Characteristics:

- Risk analysis performed
- Root causes identified
- Contingency plans created

Goal: Avoid risks whenever possible

7. Steps of Proactive Risk Management

- 1) Identify risks.
- 2) Estimate:
 - Probability
 - Impact
- 3) Rank risks.
- 4) Develop contingency plans.

Focus on:

- High probability risks
 - High impact risks
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8. Risk Identification

- Systematically **finding threats to project success.**
- Purpose: Prevent risks before they become problems.
- By identifying known and predictable risks, the project manager takes a first step toward **avoiding them when possible** and **controlling them when necessary**

Generic Risks

- **Risks common to all software projects.**
- Example: Schedule delay.

Product Specific Risks

- **Unique to a particular project.**
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9. Risk Item Checklist

A **checklist** used **to identify risks.**

Can contain:

- Questions
- Risk drivers
- Risk categories

Purpose: **Make sure no major risk is forgotten.**

10. Known & Predictable Risk Categories

Product Size Risk

- **Large systems** have **greater risk**

Business Impact Risk

- associated with constraints imposed by management or marketplace

Customer Risk

- associated with sophistication of the customer and the developer's ability to communicate with the customer in a timely manner
- Customer experienced?
- Customer cooperative?
- Requirements clear?

Process Risk

- Process defined?
- Reviews conducted?
- Quality assurance used?

Development environment

- availability and quality of the tools to be used to build the project

Technology to be built

- associated with complexity of the system
- "newness" of the technology in the system

Staff size & experience

- associated with overall technical and project experience of the software engineers who will do the work
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11. Assessing Overall Project Risk

Project managers ask questions such as:

- Is management committed?
- Are users committed?
- Are requirements understood?
- Is scope stable?
- Have customers been involved fully in the definition of requirements?
- Do end-users have realistic expectations?
- Are project requirements stable?
- Does team have required skills?
- Does the project team have experience with the technology to be implemented?
- Do all customer/user constituencies agree on the importance of the project and on the requirements for the system/product to be built?
- Is team size sufficient?

Important Rule: More "NO" answers = More project risk.

12. Risk Components

› **Performance Risk:** Will software meet requirements?

› **Cost Risk:** Will budget remain under control?

› **Support Risk:** Can software be maintained easily?

› **Schedule Risk:** Will project finish on time?

Impact Levels

Used for **risk severity**.

Level	Meaning
Negligible	Very small impact
Marginal	Minor impact
Critical	Serious impact
Catastrophic	Extremely severe

13. Risk Projection (Risk Estimation)

Determine:

1. **Probability / likelihood**
2. **Consequence**

of each risk.

Then **prioritize** risks.

› Risk Projection Steps

1. Define Probability Scale

- Establish a scale that **reflects** the **perceived likelihood** of a risk
- Example:
 - 1 = Low
 - 10 = High

2. Define Consequences

- What happens if risk occurs?

3. Estimate Impact

- Effect on:
 - Product
 - Project

4. Record Accuracy

- Document assumptions.
 - note the overall accuracy of the risk projection so that there will be no misunderstandings
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14. Risk Table

- A risk table helps **rank risks**.

Columns:

Risk	Category	Probability	Impact	RMMM
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RMMM = Risk Mitigation, Monitoring and Management

Risk Summary	Risk Category	Probability	Impact (1-4)	RMMM

Building a Risk Table

Step 1: List risks.

Step 2: Assign category.

Step 3: Estimate probability.

Step 4: Estimate impact.

- 1 = low impact on project success
- 5 = catastrophic impact on project success

Step 5: Sort highest risks first.

Step 6

- Draw cutoff line.
 - Only **highest risks get detailed attention**
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15. Risk Impact Assessment

Three things determine impact/consequences.

1) Nature

- What kind of damage occurs?

2) Scope

- How much of project is affected?

3) Timing

- When will impact occur?
- How long will it last?

Numerical: Risk Exposure Example

- **Risk identification.** Only 70 percent of the software components scheduled for reuse will, in fact, be integrated into the application. The remaining functionality will have to be custom developed.
- **Risk probability.** 80% (likely).
- **Risk impact.** 60 reusable software components were planned. If only 70 percent can be used, 18 components would have to be developed from scratch (in addition to other custom software that has been scheduled for development). Since the average component is 100 LOC and local data indicate that the software engineering cost for each LOC is \$14.00, the overall cost (impact) to develop the components would be $18 \times 100 \times 14 = \$25,200$.
- **Risk exposure.** $RE = 0.80 \times 25,200 \sim \$20,200$.

Given:

- Planned reusable components = 60
- Only 70% can actually be reused
- Risk probability = 80% = 0.80
- Average size of each component = 100 LOC
- Cost per LOC = \$14

Components that cannot be reused:

Step 1: Find Reusable Components

- $70\% \times 60 = 42$
- Reusable components = 42

Step 2: Components To Build

- $60 - 42 = 18$
- Need to develop: 18 components

Step 3: Development Cost

cost to develop these 18 components from scratch:

$$\begin{aligned} 18 \times 100 \times 14 \\ = 25,200 \end{aligned}$$

So, **Risk Impact = \$25,200**

Step 4: Risk Exposure

$$RE = Probability \times Impact$$

$$RE = 0.80 \times 25,200$$

$$RE = 20,160$$

$\approx \$20,200$

- Components to be developed = **18**
 - Risk Impact = **\$25,200**
 - Risk Exposure (RE) = **\$20,160 $\approx$ \$20,200**
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16. Risk Mitigation, Monitoring and Management (RMMM)

Risk Mitigation: Prevent risk from occurring.

Risk Monitoring: Watch risk indicators continuously.

Risk Management: Execute **contingency plan** if risk occurs.

Risks can occur after the software has been delivered to the user

Staff Turnover Mitigation Strategy

If employees may leave:

Before Project

- Find reasons for turnover
- Improve working conditions
- Improve salary if possible

During Project

- Share knowledge
- Create documentation
- Conduct peer reviews
- Assign backups

Benefits: **Project continues even if someone leaves.**

Important Exam Point

If:

$$RE < \text{Mitigation Cost}$$

Then:

-  Don't spend money mitigating.
-  Just **monitor risk**.

Because prevention costs more than damage.

17. RMMM Plan

A **document** describing:

- **Risks**
- **Mitigation**
- **Monitoring**
- **Management actions**

Objectives:

- 1) Check whether risks occur.
- 2) Ensure mitigation actions (risk aversion steps) are applied.
- 3) Collect data for future risk analysis.**

Format:

Risk information sheet			
Risk ID: P02-4-32	Date: 5/9/09	Prob: 80%	Impact: high
Description: Only 70 percent of the software components scheduled for reuse will, in fact, be integrated into the application. The remaining functionality will have to be custom developed.			
Refinement/context: Subcondition 1: Certain reusable components were developed by a third party with no knowledge of internal design standards. Subcondition 2: The design standard for component interfaces has not been solidified and may not conform to certain existing reusable components. Subcondition 3: Certain reusable components have been implemented in a language that is not supported on the target environment.			
Mitigation/monitoring: 1. Contact third party to determine conformance with design standards. 2. Press for interface standards completion; consider component structure when deciding on interface protocol. 3. Check to determine number of components in subcondition 3 category; check to determine if language support can be acquired.			
Management/contingency plan/trigger: RE computed to be \$20,200. Allocate this amount within project contingency cost. Develop revised schedule assuming that 18 additional components will have to be custom built; allocate staff accordingly. Trigger: Mitigation steps unproductive as of 7/1/09.			
Current status: 5/12/09: Mitigation steps initiated.			
Originator: D. Gagne		Assigned: B. Laster	

Example: Recording Risk Information

Project: Embedded software for XYZ system
Risk type: schedule risk
Priority (1 low ... 5 critical): 4
Risk factor: Project completion will depend on tests which require hardware component under development. Hardware component delivery may be delayed
Probability: 60 %
Impact: Project completion will be delayed for each day that hardware is unavailable for use in software testing
Monitoring approach:
 Scheduled milestone reviews with hardware group
Contingency plan:
 Modification of testing strategy to accommodate delay using software simulation
Estimated resources: 6 additional person months beginning in July

Risk due to...

Risk Due to Product Size

Attributes that affect risk:

- estimated size of the product in LOC or FP?
- estimated size of product in number of programs, files, transactions?
- percentage deviation in size of product from average for previous products?
- size of database created or used by the product?
- number of users of the product?
- number of projected changes to the requirements for the product? before delivery? after delivery?
- amount of reused software?

Risk Due to Business Impact

Attributes that affect risk:

- affect of this product on company revenue?
- visibility of this product by senior management?
- reasonableness of delivery deadline?
- number of customers who will use this product
- interoperability constraints
- sophistication of end users?
- amount and quality of product documentation that must be produced and delivered to the customer?
- governmental constraints
- costs associated with late delivery?
- costs associated with a defective product?

Risks Due to the Customer

Questions that must be answered:

- Have you worked with the customer in the past?
- Does the customer have a solid idea of requirements?
- Has the customer agreed to spend time with you?
- Is the customer willing to participate in reviews?
- Is the customer technically sophisticated?
- Is the customer willing to let your people do their job—that is, will the customer resist looking over your shoulder during technically detailed work?
- Does the customer understand the software engineering process?

Risks Due to Process Maturity

Questions that must be answered:

- Have you established a common process framework?
- Is it followed by project teams?
- Do you have management support for software engineering
- Do you have a proactive approach to SQA?
- Do you conduct formal technical reviews?
- Are CASE tools used for analysis, design and testing?
- Are the tools integrated with one another?
- Have document formats been established?

Technology Risks

Questions that must be answered:

- Is the technology new to your organization?
- Are new algorithms, I/O technology required?
- Is new or unproven hardware involved?
- Does the application interface with new software?
- Is a specialized user interface required?
- Is the application radically different?
- Are you using new software engineering methods?
- Are you using unconventional software development methods, such as formal methods, AI-based approaches, artificial neural networks?
- Are there significant performance constraints?
- Is there doubt the functionality requested is "do-able?"

Staff/People Risks

Questions that must be answered:

- Are the best people available?
 - Does staff have the right skills?
 - Are enough people available?
 - Are staff committed for entire duration?
 - Will some people work part time?
 - Do staff have the right expectations?
 - Have staff received necessary training?
 - Will turnover among staff be low?
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B. You are leading a team working on a Smart City Traffic Monitoring System. The system uses IoT sensors to collect real-time traffic data, sends it to a central server, and processes the data to update digital signboards and a mobile app with live traffic conditions. The system must integrate with multiple external services (e.g., weather updates, public transportation schedules) and comply with government security regulations. The project is on a tight deadline and involves collaboration between local developers and an outsourced team. Recently, you found out that the cloud service provider may increase its rates unexpectedly. One senior developer plans to take a long leave soon. The government authority hasn't finalized the privacy compliance policies, and the outsourced team is still learning the deployment process. There's also been talk of citywide protests that may delay some hardware installations. From the above scenario:

- I. Identify five distinct risks present in the project. For each risk, classify it as either a Project Risk, Technical Risk, or Business Risk. 5
- II. Briefly justify your classification. 5
- III. The hardware installation caused by citywide protests has a high probability of occurrence, rated at 80%. And the impact of this delay is also considered high, with an impact score of 4 on a 5-point scale. The estimated cost incurred for each day of missed installation is Rs. 4500. Using the formula for risk exposure, calculate the total estimated risk cost. 5

Risk Identified	Category	Justification
1. Cloud service provider may increase pricing unexpectedly	Business Risk/project risk	Affects operational cost and long-term budget.
2. Senior developer taking long leave	Project Risk	Affects team velocity and delivery timeline.
3. Privacy compliance policies not finalized	Technical Risk	Impacts design implementation and rework if laws change.
4. Outsourced team unfamiliar with deployment process	Technical Risk	Delays in deployment/testing can arise due to onboarding or integration errors.
5. Citywide protests may delay hardware installations	Project Risk	May postpone site readiness and incur extra labor/time charges.

III. Risk Exposure Calculation

Given:

- Probability = 80% = 0.8
- Impact score = 4
- Cost per day of missed installation = Rs. 4500

Risk Impact:

$$Impact = 4 \times 4500 = Rs. 18,000$$

Risk Exposure (RE):

$$RE = Probability \times Impact$$

$$RE = 0.8 \times 18,000$$

$$RE = Rs. 14,400$$

Answer: The estimated risk cost (Risk Exposure) is Rs. 14,400.